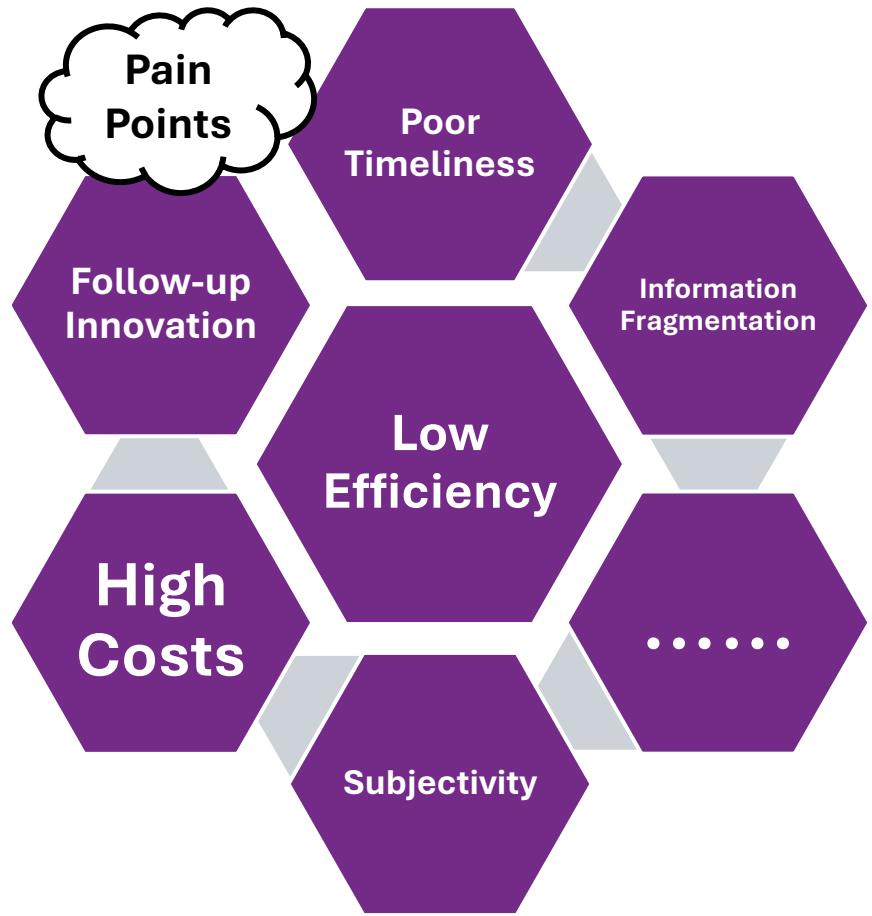
AI for technology scouting

Yu Xiao (GDE-SVVNP)
14.10.2025

AI for technology scouting

How Did We Conduct Innovative Technology Scouting Before Entering the AI Era?

- Participating in industry exhibitions and seminars
- Reading industry reports and professional literature
- Building industry networks
- Monitoring competitors' dynamics
- Consulting with universities and research institutions
- Collecting customer feedback



AI for technology scouting

Quick Scan and Make Technology Hype Cycle with the Help of AI Tools

Inputs

Step1: Scouting Area Definition

- Summarize existing technology
- Clarify the retrieval territory

Step2: Divergent ideation

- Exhaustive search
- Cross-Field
- TRIZ
- Blue-Ocean
- SCAMPER

Objective	Best Mind Tool
Solving technical challenges (e.g., airflow, noise, efficiency)	**TRIZ**, CFD Simulation
User-centered innovation (enhancing usability, experience)	**Design Thinking**, Value Proposition Canvas, QFD
Modifying existing appliances for better performance	SCAMPER, Morphological Analysis
Finding new market opportunities	**Blue Ocean Strategy (BOS /ERRC)**
Generating new ideas quickly	Brainstorming, SCAMPER

GPT4o: Best all-around tool for home appliance innovation: Design Thinking + TRIZ
GPT4o: Best for disruptive ideas: Blue Ocean Strategy

Step3: Systematic evaluation

- Technology maturity
- Feasibility
- Cost
- Industry trends
- Potential launch time

AI Tool

deepseek

豆包

Qwen3-Max

Outputs

A list covering the technologies already applied by ourselves and competitors.

A list including various fields and categories such as materials, structures, electronics, and emerging cutting-edge technologies.

AI for technology scouting

An Example of Making Noise Reduction Hype Cycle

1	Adopt a silent motor
2	Optimize the impeller design
3	Passive noise reduction with sound-absorbing cotton
4	Optimize the air duct shape
5	Install shock-absorbing pads
6	Reduce the fan speed
7	Active noise reduction

Noise reduction on ventilation.xlsx

There are some common range hood noise reduction technologies in the table.
Summarize the various technologies used in range hoods on the market.
Help me to find more technical solutions.
Add to the table.

极速 思考

1	Adopt a silent motor
2	Optimize the impeller design
3	Passive noise reduction with sound-absorbing cotton
4	Optimize the air duct shape
5	Install shock-absorbing pads
6	Reduce the fan speed
7	Active noise reduction
8	Aerodynamic design of the casing
9	Variable - frequency control optimization
10	Noise - isolation materials in the cabinet
11	Optimized fan blade surface treatment
12	Vibration - damping structure design

deepseek

豆包

Qwen3-Max

Noise reduction on ventilation.xlsx

Come up with 100 more cutting-edge technologies that have the potential to be applied to range hood noise reduction.
Organize the above into a table for me, and exclude the solutions that are completely not mass-producible.

极速 思考

Range_Hood_Noise_Reduction_Technologies.xlsx

Analyze each item in the table one by one in the form of a table from the dimensions of cost (1-5 points, 1 = lowest, 5 = highest), technology maturity (1-5 points, 1 = lowest, 5 = highest), industry trend (1-5 points, 1 = lowest, 5 = highest), and feasibility (1-5 points, 1 = lowest, 5 = highest), list the analysis results in detail, and evaluate the approximate number of years required for each solution to be potentially mass-produced and applied in range hoods. Summarize according to the categories: less than 1 year, 1-2 years, 2-5 years, more than 5 years, and completely unfeasible.

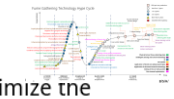
极速 思考

AI for technology scouting

An Example of Making Noise Reduction Hype Cycle

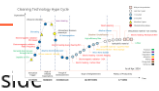
Priority list of Ventilation

Consumer needs



1. Good **fume gathering** ability to minimize the spreading of cooking fumes to other areas of the kitchen.

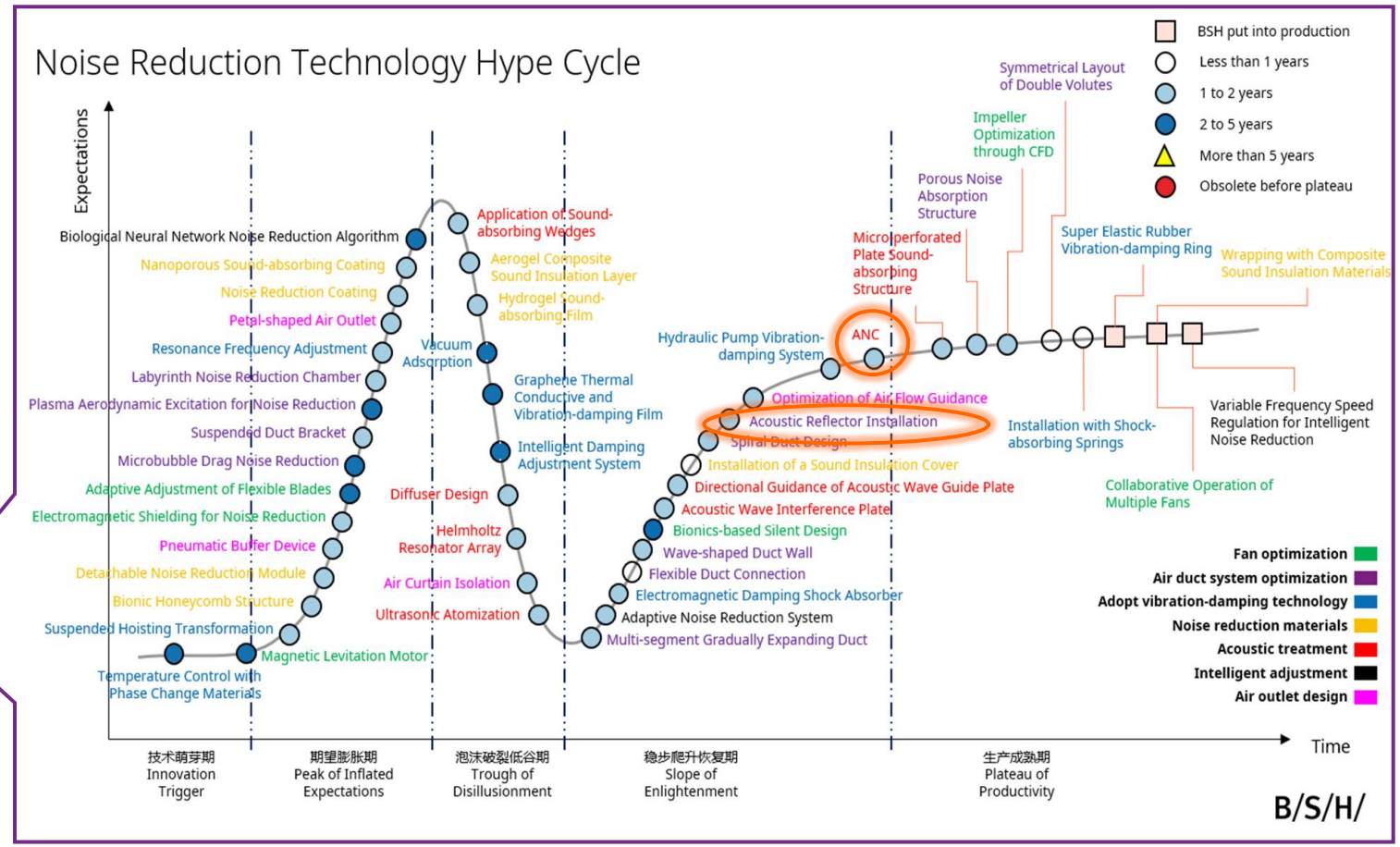
2. Stronger suction power, and intuitively it is linked with **higher air volume.**



3. Easy cleaning for both outside and inside

4. Low noise, especially on boost mode

5. Easy to use. UI is appealing and intuitive, operation is simple and easy, especially for premium and above tier.



AI for technology scouting

Benefits With AI Tool

- **Multiple times**

  increase

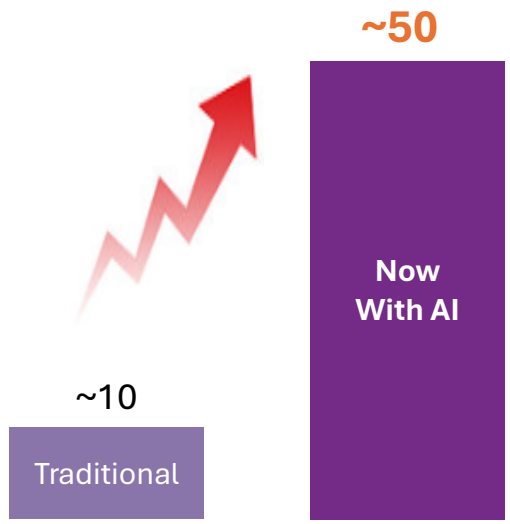
  of technical solution QTY.

- **90% decrease**

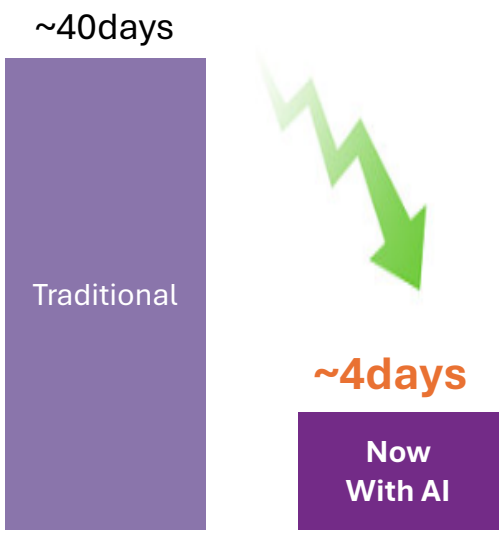
  of retrieval time.

- **0** retrieval cost

  if you'd like.



Technical Solution QTY.



Retrieval Time